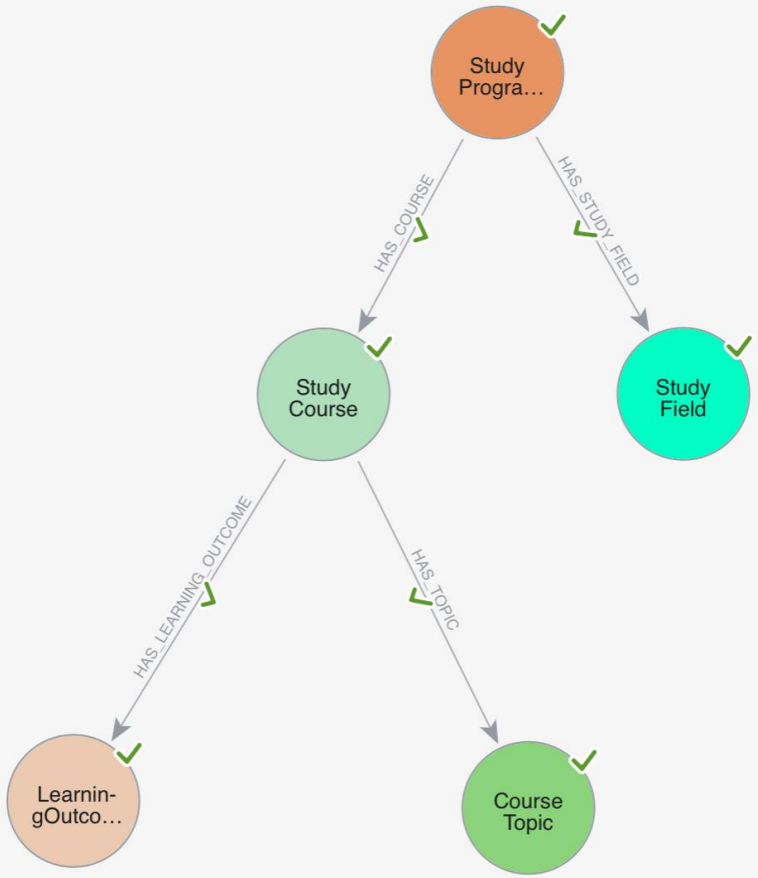
AURADB KNOWLEDGE GRAPH CREATION

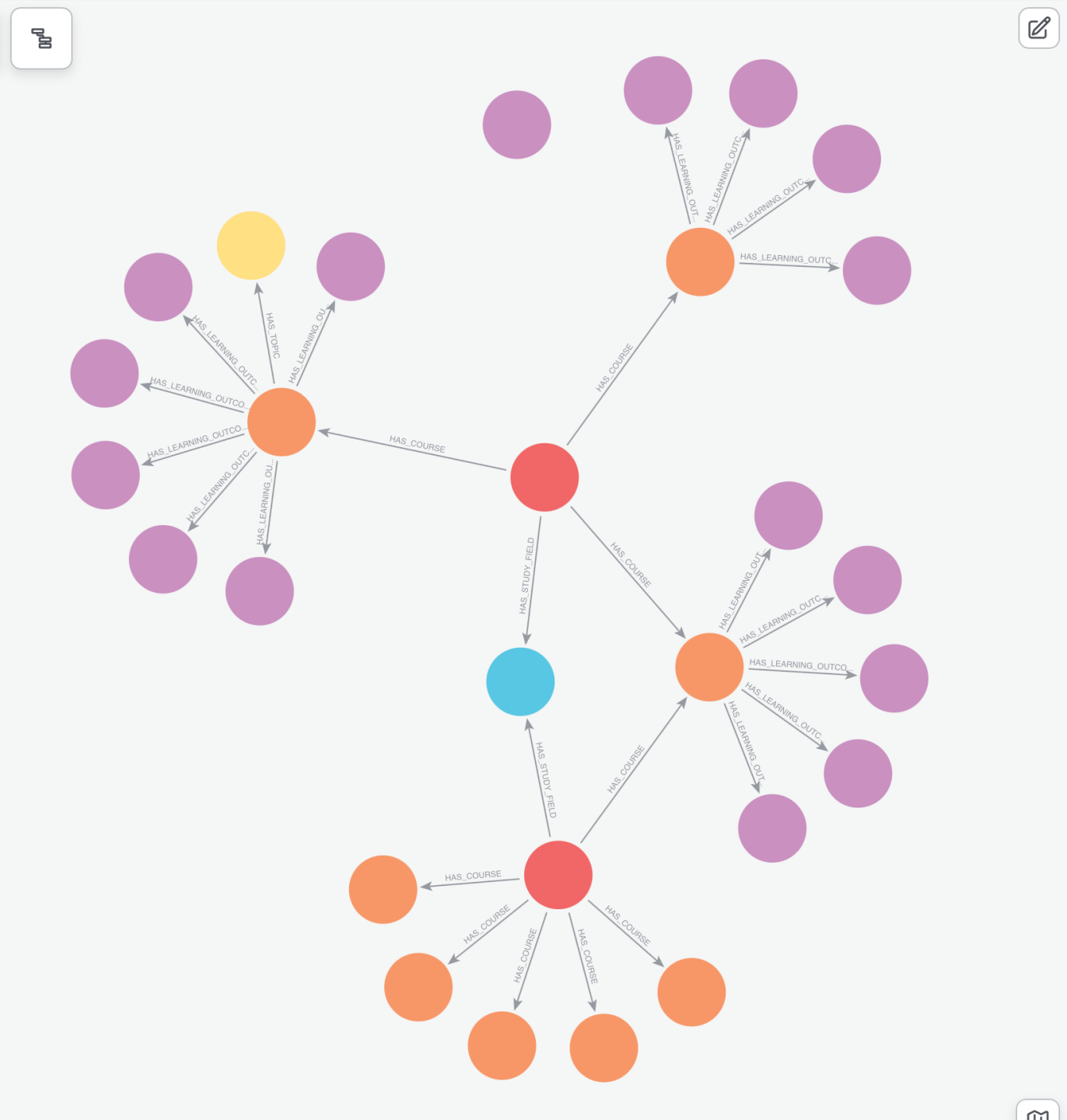
Anna Elizabete Pildava

DATA IMPORT AND SCHEMA

Data source Local Browse

- course+learning_outcome.csv
- course+topic.csv
- nodes_course_topics.csv
- nodes_learning_outcomes.csv
- nodes_study_courses.csv
- nodes_study_fields.csv
- nodes_study_programs.csv
- program+course.csv
- program+study_field.csv





Nodes | Relationships

Filter categories

All | In Scene | Off Scene

CourseTopic	1
LearningOutcome	16
StudyCourse	8
StudyField	1
StudyProgram	2

QUERY 1 – ALL COURSES WITH ALL OUTCOMES

Stream | Course outcomes | DE0745 | BI 6 credit points | matching courses

```
1 MATCH (c:StudyCourse)-[:HAS_LEARNING_OUTCOME]->(lo:LearningOutcome)
2 RETURN
3   c.courseId AS CourseId,
4   c.title AS CourseTitle,
5   lo.text AS LearningOutcome
```

Run

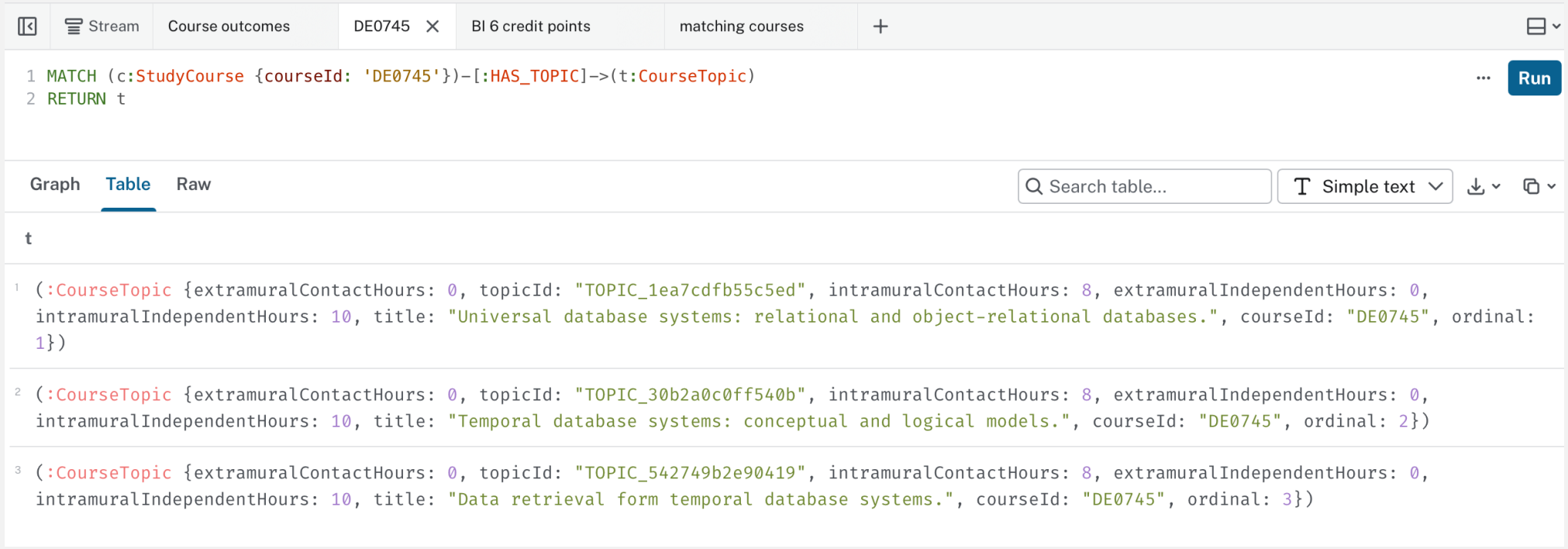
Table | Raw

Search table... | Graph chip

	CourseId	CourseTitle	LearningOutcome
1	"AD0188"	"Architectural Morphology in Digital Humanities"	"Orients in the diversity of influences and manifestations of the art of creating an environment in a cultural environment."
2	"AD0188"	"Architectural Morphology in Digital Humanities"	"Is able to explain the causes of current issues in the formation of the spatial environment in society and predict the course of process development."
3	"AD0188"	"Architectural Morphology in Digital Humanities"	"Is able to perceive and critically process information, evaluate the reliability of its sources and the degree of compliance with the set goal (for the task)."
4	"AD0188"	"Architectural Morphology in Digital Humanities"	"Is able to logically structure an idea, choose adequate argumentation, engage in discussion, oppose."
5	"AD0188"	"Architectural Morphology in Digital Humanities"	"Is able to argue the purposefulness of the resources and methods used in the specific environmental development plan or scientific research and their compliance with the set goal (task)."
6	"AD0188"	"Architectural Morphology in Digital Humanities"	"Is able to present the results of the work, choosing the most purposeful means of verbal and visual expression."
7	"AD0188"	"Architectural Morphology in Digital Humanities"	"Is able to present the results of the work and defend them in an academic discussion."

Started streaming 366 records after 18 ms and completed after 20 ms.

QUERY 2 – DE0745 STUDY COURSE TOPICS

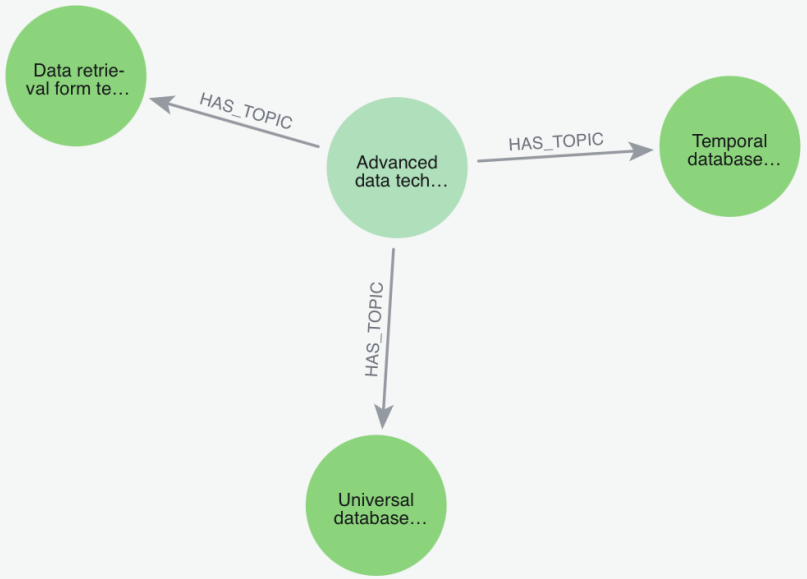


QUERY 2 – DE0745 STUDY COURSE TOPICS

Stream | Course outcomes | DE0745 | BI 6 credit points | matching courses

```
1 MATCH (c:StudyCourse {courseId: 'DE0745'})-[:HAS_TOPIC]->(t:CourseTopic)
2 RETURN t
```

Graph | Table | Raw



QUERY 3 – BI COURSES WITH 6 CREDIT POINTS

Stream | Course outcomes | DE0745 | BI 6 credit points | matching courses

```
MATCH
  (p:StudyProgram {title: 'Business Informatics'})-[:HAS_COURSE]->
  (c:StudyCourse)
WHERE c.creditPoints = 6.0
RETURN c.title
```

Table | Raw

c.title
"Advanced data technologies"
"Enterprise Architecture and Requirements Engineering"
"Systems Theory"
"Business Process Management and Engineering"
"e-Business Solutions"
"Portfolio Management Technologies"
"Knowledge Management Systems"
"Service Science, Management, and Engineering"
"Quality, Risk and Security Technologies"
"Artificial Intelligence in Business"
"Information Systems Security Engineering"
"Enterprise Information Technology Architecture, Applications and Integration"
"Digital Transformation"
"Business Analytics"
"Business Analytics with Machine Learning"
"Business Law"
"Business Ethics"
"Entrepreneurship"
"Business Communication Skills"
"Entrepreneurial Finance"
"Leadership"
"Corporate Governance"
"Pedagogical Process. Basics"

QUERY 4 – COURSES THAT BOTH STUDY PROGRAMS HAVE

Stream | Course outcomes | DE0745 | BI 6 credit points | matching courses

```
1 MATCH
2   (p1:StudyProgram {programId: 'DMB0'})-[:HAS_COURSE]->
3   (c:StudyCourse)<-[:HAS_COURSE]-
4   (p2:StudyProgram {programId: 'HMD0'})
5 RETURN c.title
```

Table | Raw

c.title
"e-Business Solutions"
"Portfolio Management Technologies"
"Knowledge Management Systems"
"Communication and Presentation Skills"